

Assignment 1: Scenario-Based AIMS Implementation

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Organisation: AutoDrive Insurance

AI System: ClaimVision

Total Marks: 100

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Task 1: Define the Scope (Clause 4.3)

AIMS Scope Statement

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1. Purpose

This Scope Statement defines the boundaries, applicability, inclusions, and exclusions of AutoDrive Insurance's Artificial Intelligence Management System (AIMS), established in accordance with ISO/IEC 42001:2023.

2. Organisational scope

AutoDrive Insurance acts as both the **AI Producer** (the ClaimVision model is developed entirely in-house by five data scientists) and the **AI Provider/Operator** (it deploys and operates the system through its customer-facing mobile application). The AIMS, therefore covers both roles across the full life cycle of the system.

3. Systems and processes in scope

- The **ClaimVision AI model**:- the computer-vision damage-assessment and repair-cost-estimation model.
- The **mobile app feature**:- photo capture, submission, and results interface.
- The **automated payout decision engine**:- autonomous approval of payouts under USD 2,000.
- The **escalation workflow**:- automatic flagging of payouts at or above USD 2,000 to human adjusters.
- **Supporting data infrastructure**:- the 500,000-photo training-data repository, data pipelines, labelling, and the model training/retraining environment.

- **Monitoring and logging systems:-** model performance, decision logs, drift detection, and audit trails.

4. Departments and functions in scope

- Data Science / AI Engineering (the five-person in-house team)
- Claims Operations (human adjusters receiving escalations)
- IT / MLOps (deployment, infrastructure, monitoring)
- Compliance, Risk, and Legal (governance, impact assessment, regulatory assurance)
- Customer Support (complaint and appeal handling)
- Board and executive management (governance, resourcing, oversight)

5. Life cycle stages covered

Concept and business case → design and development → data acquisition and preparation → verification and validation → deployment → operation and monitoring → continuous learning/retraining → retirement and decommissioning (as applicable).

6. External interfaces

The **third-party data broker** from whom the training dataset was purchased is an external supplier. AutoDrive's management of this relationship (procurement, due diligence, data provenance) is within the AIMS; the broker's own internal operations sit outside the boundary.

7. Exclusions and justification

1. **All non-AI products and processes** (e.g., core underwriting, pricing, non-ClaimVision claims handling, and general insurance operations); excluded because the AIMS is deliberately scoped to the ClaimVision AI system, keeping certification focused and proportionate to the highest-risk AI-enabled process.
2. **Human adjuster professional judgement on escalated claims (≥ USD 2,000)**; excluded; only the AI's flagging and handoff of the claim is in scope. Human professional discretion is not an "AI system" and therefore falls outside an AIMS.
3. **Third-party data broker's internal operations** excluded; these are outside AutoDrive's control and are managed through supplier controls rather than certified as part of AutoDrive's AIMS.
4. **No clause of ISO/IEC 42001 is excluded.** Given the single-system, single-organisation scope, all requirements of Clauses 4-10 are applicable.

8. Interested parties (context)

Customers (claimants), the Board and regulators, employees (data scientists and adjusters), the third-party data broker, and the public. Their relevant requirements; fairness, accuracy, transparency, data protection, and regulatory assurance; frame the AIMS's boundaries and objectives.

Task 2: Preliminary AI Risk Assessment

The assessment below applies the impact-assessment and risk-assessment approach of ISO/IEC 42001 Clause 6.1. All Annex A references are to ISO/IEC 42001:2023.

Risk 1: Algorithmic bias / disparate accuracy across vehicle attributes

Risk scenario

The training set of 500,000 historical photos is not demonstrated to be representative of the real-world fleet. If certain vehicle colours, models, or body types are over- or under-represented, the model may systematically misestimate repair costs for specific groups, for example, recognising damage more accurately on common colours (silver, black, white) than on less common colours, or on older models than on newer designs. This creates differential, unfair outcomes even where aggregate accuracy looks acceptable.

Potential impact

On the **customer**, under-estimated repairs leave affected claimants underpaid and facing out-of-pocket costs, amounting to unfair and potentially discriminatory treatment. On the **organisation**, systematic over- or under-payment causes financial loss, regulatory scrutiny, reputational damage, and loss of customer trust.

Annex A controls to treat the risk

- **A.5.2 AI system impact assessment process** assess fairness impacts on affected groups before deployment.
- **A.6.2 AI system development** embed fairness requirements and bias mitigation in the design and development process.
- **A.6.5 AI system verification and validation** validate performance across vehicle subgroups, not only on aggregate accuracy.
- **A.7.1 Data acquisition** acquire and curate a representative, balanced dataset with documented coverage across colours, models, and body types.
- **A.8.1 Transparency of the AI system** disclose system limitations and the availability of human review.

Preliminary risk level: High.

Risk 2 Automation error from out-of-distribution inputs

Risk scenario

The system autonomously approves payouts under USD 2,000 with no human intervention. When the model encounters inputs outside its trained domain, a newly released vehicle model, novel materials, or unusual damage patterns, it can produce confident but wrong estimates (e.g., failing to detect expensive sensors). Because the decision path is automated, the error propagates directly to a payout decision with no human check.

Potential impact

On the **customer**, incorrect payouts cause underpayment (leaving the claimant out of pocket) or overpayment (creating later claw-back disputes), eroding trust in automated decisions. On the **organisation**, the exposure includes financial loss, viral reputational damage, regulatory scrutiny of autonomous decision-making, and fraud exposure from over-generous estimates.

Annex A controls to treat the risk

- **A.6.5 AI system verification and validation;** test the model against out-of-distribution and "unseen vehicle" scenarios before release.
- **A.9.2 Human oversight;** require human review for low-confidence predictions or unrecognised vehicle types, even below the USD 2,000 threshold.
- **A.6.6 AI system continuous learning;** monitor for drift and trigger retraining when model coverage no longer matches the real-world fleet.
- **A.6.7 AI system operation and monitoring;** log decisions and monitor error rates and confidence distributions in production.

Preliminary risk level: High.

Risk 3 Third-party data provenance and quality (data broker)

Risk scenario

The 500,000 training photos were purchased from a third-party data broker with no documented assurance of provenance, legal basis, consent, labelling accuracy, or representativeness. The data may contain mislabelled damage, duplicates, or low-quality images, may carry unresolved licensing or data-protection obligations, and may embed sampling bias into the model ("garbage in, garbage out").

Potential impact

On the **organisation**, the exposure includes legal and regulatory risk (data protection and IP/licensing), model-quality defects, and a training-data supply chain that cannot be defended during certification or regulator review.

Customers cannot challenge or understand payout decisions, leading to claims of bad-faith insurance practices and non-compliance with consumer transparency requirements.

Annex A Controls:

- **A.8.1 (Transparency of the AI system):** Provide clear notifications in the mobile app detailing how the AI processes photos, what factors influence estimates, and how customers can request human review.
- **A.8.4 (Explainability of the AI system):** Implement explainable AI (XAI) techniques (e.g., heatmaps/Grad-CAM overlays) showing customers and adjusters which photo regions dictated the cost assessment.

Task 3: Statement of Applicability (SoA)

Control ID	Control Description	Included? (Y/N)	Justification
A.6.2	AI system development	Y	Included. AutoDrive acts as the AI Producer, developing ClaimVision entirely in-house with a team of 5 data scientists. Comprehensive software development lifecycle controls, security standards, and validation protocols are mandatory.
A.6.6	AI system continuous learning	N	Excluded. ClaimVision uses a static, pre-trained model deployed in production to ensure deterministic payout evaluation. It does not perform real-time, autonomous online updates on production data, eliminating risks of runtime concept drift.

A.7.1	Data acquisition	Y	Included. The model relies heavily on 500,000 third-party historical photos purchased from a data broker. Controls are required to verify intellectual property rights, data provenance, privacy consent, and dataset integrity.
A.8.1	Transparency of the AI system	Y	Included. The system directly impacts end consumers by automatically deciding financial payouts. Transparency controls are critical to inform users that an automated decision is occurring and to outline dispute mechanisms.

Task 4: Corrective Action Plan (Clause 10.1)

1. React (Immediate Containment)

- **Immediate Customer Containment:** Contact the affected customer directly, apologize for the discrepancy, pause automated processing on their claim, assign a senior human adjuster to re-evaluate the EV vehicle manually, and issue the correct \$2,500 payout immediately.
- **Systemic Operational Containment:** Temporarily adjust the ClaimVision confidence threshold. Immediately route all claims involving electric vehicles (EVs), hybrid models, or vehicle model years within the last 18 months to human adjusters for manual review, preventing further automated miscalculations.
- **Communication:** Issue a measured public response stating that the claim was re-evaluated by an adjuster, corrected, and that a feature update is underway to enhance new-vehicle model recognition.

2. Root Cause Analysis (RCA)

- **Failed Process:** Inadequate Data Lifecycle & AI Lifecycle Management (Clause 6.1 / Annex A.7).

Detailed Breakdown:

Data Representation Gap: The third-party dataset of 500,000 historical photos was static and lacked sufficient representation of recently released EV models featuring integrated bumper sensors (ADARS/LiDAR/ultrasonic units).

- **Verification Deficit:** The verification and validation procedures failed to test model performance against a diverse matrix of newly released vehicle architectures prior to production release.

3. Corrective Action (Systemic Change)

- **Trigger-Based Model Retraining & Dataset Updates:** Establish a mandatory quarterly data refresh protocol that ingests OEM (Original Equipment Manufacturer) repair data, new vehicle schematics, and sensor layout changes.
- **Pre-Deployment Coverage Testing (Change Management):** Update the AIMS verification policy (A.6.2/A.6.4) to mandate that all new model-year vehicles undergo explicit out-of-distribution (OOD) testing before being authorized for straight-through AI payout processing.
- **Dynamic Confidence Scoring:** Implement automated fallback logic where the AI system automatically routes a claim to a human adjuster whenever the model detects a vehicle model year or body design with a feature-confidence score below 95%.